



The Driver Behaviour Questionnaire as a predictor of accidents: A meta-analysis[☆]

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ABSTRACT

Introduction: Through a meta-analysis, this study investigated the relation of errors and violations from the Driver Behaviour Questionnaire (DBQ) to accident involvement. **Method:** We identified 174 studies using the DBQ, and a correlation of self-reported accidents with errors could be established in 32 samples and with violations in 42 samples. **Results:** The results showed that violations predicted accidents with an overall correlation of .13 when based on zero-order effects reported in tabular form, and with an overall correlation of .07 for effects reported in multivariate analysis, in tables reporting only significant effects, or in the text of a study. Errors predicted accidents with overall correlations of .10 and .06, respectively. The meta-analysis also showed that errors and violations correlated negatively with age and positively with exposure, and that males reported fewer errors and more violations than females. Supplementary analyses were conducted focusing on the moderating role of age, and on predicting accidents prospectively and retrospectively. Potential sources of bias are discussed, such as publication bias, measurement error, and consistency motif. **Impact on Industry:** The DBQ is a prominent measurement scale to examine drivers' self-reported aberrant behaviors. The present study provides information about the validity of the DBQ and therefore has strong relevance for researchers and road safety practitioners who seek to obtain insight into driving behaviors of a population of interest.

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1. Introduction

It has long been documented that safe driving is not only accomplished by being able to drive in a relatively error-free manner. Intentional violations and risk taking are important determinants of road safety as well (Jonah, 1986; Robertson & Baker, 1975; Schuman, Pelz, Ehrlich, & Selzer, 1967). In 1990 Reason, Manstead, Stradling, Baxter, and Campbell introduced the Driver Behaviour Questionnaire (DBQ), which consisted of 50 items describing a variety of errors and violations during driving. Respondents had to indicate how often each aberration occurred to them during the last year on a scale between 0 (*never*) to 5 (*nearly all the time*). By conducting a principal component analysis on the results of 520 drivers, Reason et al. showed that errors were statistically distinct from violations, supporting the hypothesis that errors and violations are governed by different psychological mechanisms. Errors reflect performance limits of the driver such as those related to perceptual, attentional, and information processing abilities. Violations represent the style in which the driver chooses to drive and habits established after years of driving. The distinction

between errors and violations is analogous to the distinction between driver performance and driver behavior (Evans, 2004), skills and safety motives (Lajunen & Summala, 1995), and driving skill and driving style (Elander, West, & French, 1993).

Since the work of Reason et al. (1990) the popularity of the DBQ has increased tremendously. Currently, at least 174 studies exist that have used the DBQ or a modified version (Fig. 1). The distinction between errors and violations has been found cross-culturally (Lajunen, Parker, & Summala, 2004; Özkan, Lajunen, Chliaoutakis, Parker, & Summala, 2006) and in special groups such as professional drivers, motor riders, traffic offenders, probationary drivers, parent-child pairs, young women, and older drivers (Bianchi & Summala, 2004; Brookland, Begg, Langley, & Ameratunga, 2008; Conner & Lai, 2005; Dobson, Brown, & Ball, 1998; Freeman, Wishart, Davey, Rowland, & Williams, 2009; Rimmö & Hakamies-Blomqvist, 2002; Schwebel et al., 2007; Steg & Van Brussel, 2009; Stevenson, Palamara, Morrison, & Ryan, 2001; Sullman, Meadows, & Pajo, 2002; Wallace, 2008). Furthermore, the DBQ errors and violations factors are strongly situated in a network of correlations with other questionnaires and tests such as Trait Anxiety, Driving Style Questionnaire, Big Five personality factors, Driver Attitude Questionnaire, Decision Making Questionnaire, Cognitive Failures Questionnaire, Propensity for Angry Driving Scale, Driver Perception of Pressure, Driving and Riding Avoidance Scale, Religious Orientation Scale, and Sensation Seeking Scale (Chapman, Ismail, & Underwood, 1999; Conner & Lai, 2005; Dobson et al., 1998; Lucidi et al., 2010; Maxwell, Grant, & Lipkin, 2005; Parker, Stradling, & Manstead, 1996; Schwebel, Severson, Ball, & Rizzo,

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2006; Shahar, 2009; Sümer, Lajunen, & Özkan, 2005; Taylor & Sullman, 2009; Van de Sande, 2008; Yildirim, 2007).

One of the most important applications of the DBQ is the prediction of individual differences in accident involvement. However, it is currently unclear to what extent the DBQ can predict accident involvement, because the results in the literature appear to be heterogeneous. For example, Freeman et al. (2009) and Sümer (2003) reported positive correlations (.16 and .18, respectively) between errors and accidents, whereas Stephens and Groeger (2009) found a negative correlation between lapses and accidents (−.16). Özkan and Lajunen (2005a) reported a correlation between ordinary violations and accidents of .35, whereas others found insignificant correlations, even pointing to the opposite direction (e.g., −.04 for highway code violations in Davey, Wishart, Freeman, & Watson, 2007). A review of Stradling, Parker, Lajunen, Meadows, and Xie (1998) stated that violations, not errors, predicted accidents. DeLucia, Bleckley, Meyer, and Bush (2003), on the other hand, found that errors, not violations, predicted accidents, and according to Blokey and Hartley (1995) neither errors nor violations were significant accident predictors. More recently, af Wählberg, Dorn, and Kline (in press) observed that in the literature “errors and lapses, taken together, have been significant predictors of accidents about as many times as the various violation factors” (p. 12).

Schmidt (1992) and Gardner and Altman (1986) explained that individual studies contain only little information and that apparent inconsistencies in the effect sizes and *p*-values can usually be explained through sampling error alone. A meta-analysis is essential to cope with sampling error, to expose the consistency of the effects, and to aid in theory forming (Schmidt, 1992). This study used a meta-analysis to estimate the predictive correlations of the DBQ errors and violations factors with regard to self-reported and registered accidents. The effects of age, gender, and exposure (mileage and hours driven per week) were evaluated as well, as these variables are known confounders of accident involvement (Lourens, Viissers, & Jessurun, 1999; Massie, Campbell, & Williams, 1995).

2. Method

A literature search was conducted to recover published and unpublished studies that used the DBQ. The searches were conducted with Google Scholar, Web of Science, and Scopus, and by tracing the references of the retrieved documents.

Given the purpose of this meta-analysis to summarize as much of the empirical evidence of the DBQ-accident correlation as possible, all types of scientific studies, that is, journal articles, papers presented at scientific conferences, book chapters, reports, PhD dissertations, and Bachelor or Master theses, were eligible for inclusion. There exist many variants of the DBQ, such as translations or variants that include culture-specific aberrations (e.g., Xie & Parker, 2002), with diverse

numbers of items (from 10 in Rowland, Davey, Freeman, & Wishart, 2009 up to 112 in Kontogiannis, Kossivelou, & Marmaras, 2002), and different numbers of extracted factors (from one in e.g., Hennessy & Wiesenthal, 2005 to seven in Kontogiannis et al., 2002). In our meta-analysis, all DBQ variants including those for special groups such as professional drivers and motor riders were eligible for inclusion. However, studies that extracted neither a violations factor nor an errors factor (such as Furnham & Saipie, 1993, extracting five factors related to risk taking) were excluded. We identified a number of DBQ studies in languages other than English. These were excluded as well because of limited accessibility and the practical difficulties of translation. At this stage, our literature search had retrieved 174 DBQ studies.

Next, a selection was made among the retrieved DBQ studies. First, studies on children, pedestrians, and moped riders were excluded (Díaz, 2002; Elliott & Baughan, 2004; Mann & Sullman, 2008; Steg & Van Brussel, 2009). A substantial number of studies re-analyzed DBQ data from the same sample of respondents. If data were apparently re-used, we included only the study that reported the most comprehensive information on the relation between the DBQ factors and external criteria; the other studies using the same sample were excluded. It was also possible that the same sample was analyzed more than once within the same study as part of a longitudinal research (e.g., Conner & Lai, 2005). For these studies, we only used the available data from the first measurement instance and discarded the measurements that took place at a later phase. A number of studies included analyses on more than one separate sample of respondents (af Wählberg, 2010; af Wählberg et al., in press; Bener, Özkan, & Lajunen, 2008; Dobson et al., 1998). These samples were treated as independent.

Per sample, the correlations between the DBQ factor scores and the following six criteria were noted down, if available: self-reported number of accidents, recorded number of accidents (only in af Wählberg et al., in press), gender, age, mileage, and hours driven per week. If an effect size different than a correlation was reported (e.g., means and SDs, *F*-statistics between two groups, *t*-statistics, odds ratios), the effect size was converted into a correlation by using the equations reported in Borenstein, Hedges, Higgins, and Rothstein (2009). If correlations between DBQ factors and different types of accidents were reported, such as all accidents, culpable-only accidents, and passive/active accidents, we noted down only the correlation with all accidents. If the correlation with all accidents was not reported, we noted down the average of the available DBQ-accident correlations. In some cases, the effects were reported as an *F*-statistic of multiple groups (e.g., Analysis of Variance conducted on three or more age groups). In these cases, no correlation coefficient was calculated because we deemed that insufficient information was available for that. The DBQ-accidents correlations of Meadows (1994) and Mesken, Lajunen, and Summala (2002) were excluded due to uncertain reporting or data processing and because the observed effects were considered as outliers.

A number of DBQ studies used regression analysis, path analysis, structural equation modeling, or another multivariate technique, without reporting zero-order effects. In these cases we recorded the effect size from the multivariate analysis (typically standardized path coefficients or regression coefficients). The strength of a relationship between a predictor and a criterion variable obtained in a multivariate analysis depends on which predictors the authors included and therefore differs from the zero-order correlations. Furthermore, the corresponding standard errors are often unknown. Another issue was that some analyses did not report all relevant effects but only the statistically significant ones, or used stepwise regression analysis. Stepwise regression analysis uses an automatic procedure to select the strongest predictors to include in the meta-analysis and can distort the effect sizes because multiple hypotheses are tested, capitalizing on chance (Whittingham, Stephens, Bradbury, & Freckleton, 2006). To be able to investigate the zero-order effects separately from the more

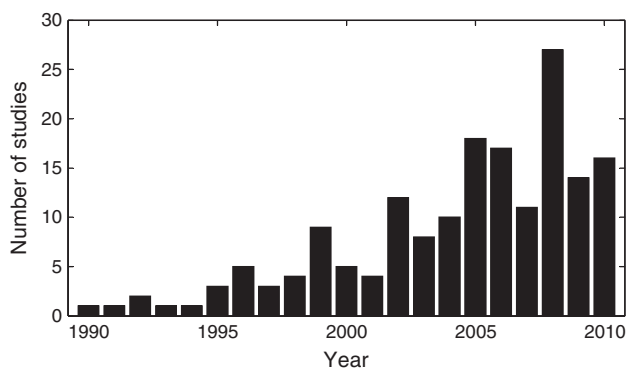


Fig. 1. Number of English-written studies per year using the original DBQ or a modified version of it.

distorted effect sizes, we coded each obtained correlation into (Z): zero-order effects reported in a table (typically a correlation matrix), and (O): other, that is, effects in multivariate analysis, in a table omitting the insignificant effects, or in the text of a study.

The DBQ factors per sample were coded into three categories: (1) errors, (2) violations, and (3) other. The following factors were classified as errors, representing an exhaustive list of the terminology used in the included studies: control errors, dangerous errors, fatigue and distractions, hazardous errors, general errors, inattention errors, inattention lapses, inexperience errors, lapses, mistakes, memory lapses, mistakes, nonhazardous errors, serious errors, skill errors, (action) slips, and traffic errors. The following factors were classified as violations: aggression-speeding, aggressive violations, anger/hostility, dangerous violations, deliberate violations, emotional violations, fast driving, highway violations, interpersonal violations, maintaining progress violations, negligence, ordinary violations, parking violations, pushing-speeding, risky violations, self-willed violations, social disregard, speeding violations, stunts, style, and traffic violations. The remaining factors were classified as others. These were, for example, factors identified in specific variants of the DBQ such as the factor seatbelt use in Şimşekoğlu and Lajunen (2009) or the factors driver distraction and pre-trip maintenance in Wills, Watson, and Biggs (2006). When correlations were available for multiple types of errors or violations, these were averaged. For example, when a correlation of .07 was described between errors and accidents and a correlation of .09 between lapses and accidents, then a correlation of .08 was noted between errors and accidents.

The meta-analysis was conducted per predictor variable (DBQ errors and DBQ violations), per criterion (number of self-reported accidents, number of recorded accidents, age, gender, mileage, and hours driven per week), and per correlation type (Z and O). We used both random- and fixed-effects models to assess overall correlation magnitude. Because both models yielded the same trends, herein only the fixed-effects results are presented.

3. Results

In total, 76 samples from 70 studies fulfilled our inclusion criteria and yielded a correlation with at least one of the six criteria. These

samples were used in the meta-analysis, the results of which are shown in Table 1. A correlation of self-reported accidents with errors was established in 32 samples and with violations in 42 samples.

The most important result of this meta-analysis was that both errors and violations correlated positively with self-reported accident involvement, with zero-order effects in tabular form being significantly stronger predictors than effects reported in multivariate analyses, in a table including only the statistically significant effects, or in the text of the studies. The correlations of errors and violations with recorded accidents were not statistically significant.

The funnel plots of the DBQ-accident relationships (Figs. 2 and 3) revealed no apparent asymmetry for the zero-order effects. For errors, we found that one of the large-sample studies (Freeman et al., 2009 involving 4,792 professional drivers) revealed a relatively large correlation with self-reported accidents, outside the 95% confidence interval. Expectedly, zero-order effects were distributed more homogeneously than the other effects. For the very small samples, some publication bias was evident, with a few correlations being considerably more positive than the overall effect ($>.3$), while there were no correlations considerably more negative than the overall effect ($<-.1$).

The meta-analysis of the relationship between the DBQ factors and gender, age, and exposure revealed that violations, and to a lesser extent errors, reduced with age, that males reported more violations and fewer errors than females, and that an increased mileage related to an increased violation score. The relationship between mileage and errors was less consistent, yielding a positive correlation for the zero-order effects and a negative correlation for the other effects. Finally, it was found that those who reported driving more hours per week reported more errors and violations, although this latter result was not equally significant for all correlations.

3.1. Supplementary analyses

A number of studies reported that DBQ violations increased with experience during the first months of licensure (Bjørnskau & Sagberg, 2005; De Craen, 2010; Wells, Tong, Sexton, Grayson, & Jones, 2008a,b), a result that seems to contradict the negative overall correlation between violations and age of our meta-analysis.

Table 1
Results of meta-analysis.

Criterion variable	Type of effect	DBQ aberration	Number of samples	Total sample size	Overall correlation	95% low	95% high
Self-reported accidents	Zero-order	Errors	23	36,281	.10	.09	.11
		Violations	30	46,338	.13	.12	.14
	Other	Errors	9	8,080	.06	.04	.08
Recorded accidents	Zero-order	Violations	12	9,798	.07	.05	.09
		Errors	1	291	.03	-.09	.14
	Other	Violations	3	659	.05	-.03	.13
		Errors	0	0	-	-	-
		Violations	0	0	-	-	-
Age	Zero-order	Errors	18	24,255	-.05	-.06	-.04
		Violations	22	25,460	-.22	-.23	-.21
	Other	Errors	8	7,380	-.06	-.08	-.04
		Violations	14	10,701	-.36	-.37	-.34
Gender (1 = Male, 2 = Female)	Zero-order	Errors	18	22,976	.04	.03	.05
		Violations	23	24,274	-.15	-.16	-.14
	Other	Errors	13	13,531	.06	.04	.07
		Violations	15	11,205	-.16	-.17	-.14
Mileage	Zero-order	Errors	13	20,127	.02	.01	.04
		Violations	14	20,397	.06	.05	.08
	Other	Errors	3	3,244	-.09	-.13	-.06
		Violations	8	6,292	.12	.10	.14
Hours driven per week	Zero-order	Errors	4	6,237	.12	.10	.14
		Violations	4	6,237	.01	-.02	.03
	Other	Errors	1	113	.13	-.06	.30
		Violations	3	575	.17	.09	.25

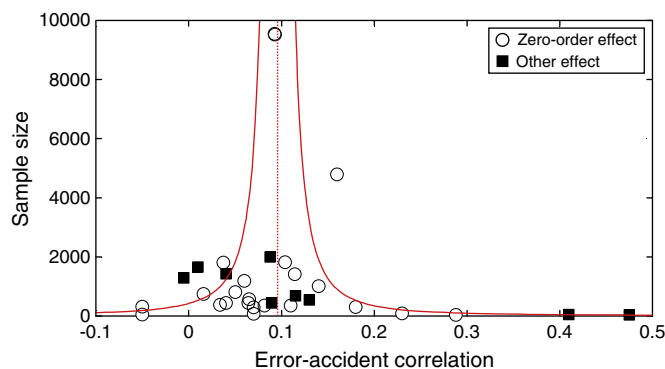


Fig. 2. Funnel plot for the errors-accident correlation. A 95% confidence interval is drawn around the overall zero-order effect. Note that the correlations from the two large-sample studies (Elliott, Sexton, & Keating, 2003; TRL, 2008) lie almost exactly on top of each other.

To obtain further information on the effects of age, we conducted a meta-regression using the mean age of the sample as independent variable, the zero-order violations-age correlation and errors-age correlation as dependent variable, and the sample size as weight. When the mean age was not reported, we estimated it based on the available data (typically from the frequency counts of stratified-by-age groups).

The meta-regression showed that the violations-age correlation was uncorrelated with the age of the sample (standardized $\beta = .10$, $n = 22$, $p = .670$), whereas the errors-age correlation increased with age (standardized $\beta = .62$, $n = 18$, $p = .006$). These effects are graphically illustrated in Fig. 4. It can be seen that for the youngest sample (i.e., Özkan & Lajunen, 2005a), violations increased with age, whereas for all other samples, violations decreased with age. Errors, on the other hand, decreased stronger with age for the younger drivers than they did for the older drivers.

Using meta-regression, we also identified a negative trend between the mean age and the violations-accident correlation (standardized $\beta = -.48$, $n = 30$, $p = .008$) but no significant relationship between the mean age and the errors-accident correlation (standardized $\beta = .09$, $n = 23$, $p = .671$).

Finally, we investigated whether the DBQ could predict self-reported accidents prospectively as well as retrospectively. We obtained data from the Transport Research Laboratory, Safety, Security, and Investigations Division [TRL] (2008), including more than 10,000 beginner drivers who completed the DBQ as part of a longitudinal study (for a full description of these data see Wells et al., 2008a). Self-reported accident and DBQ responses were collected after 6, 12, 24, and 36 months of licensure. The results are shown in Table 2 and indicate that the errors and violations factors predicted

accidents prospectively and retrospectively with rather consistent strength throughout. In some but not all cases, correlations seemed somewhat stronger when assessed within the same questionnaire as compared to correlations for time periods relatively far apart.

4. Discussion

At least 174 English-written studies have used the DBQ, but no overview existed that quantitatively synthesized the predictive correlations of the errors and violations factors. Our results showed that errors and violations are about equally strong predictors of self-reported accidents. This result differentiates from earlier studies that concluded that only errors (e.g., DeLucia et al., 2003) or only violations (see Stradling et al., 1998 for a narrative review) were significant accident predictors. We also found that age correlated negatively with violations and errors. Furthermore, violations were a stronger predictor of accidents amongst young drivers than they were amongst old drivers. This is in line with results obtained by Parker, McDonald, Rabbitt, and Sutcliffe (2000) who also recognized that violations were not a significant predictor of accidents amongst older drivers.

As regards recorded accidents, we showed that the total sample size was insufficient to derive any reasonable conclusion. That is, the confidence intervals were too wide to be able to conclude that the DBQ predicts recorded accidents or whether it is a better or worse predictor for self-reported accidents than it is for recorded accidents. It should be noted that objective (i.e., recorded) accident data are not necessarily a more valid criterion than self-reported data. Self-reported data are susceptible to various distortions (discussed below), but recorded data are subject to biases too, one of them being that certain groups of drivers may be overrepresented for reasons unrelated to their accident involvement. For example, given a crash, older drivers are more likely to suffer an injury than younger drivers, and consequently, accidents of older drivers are more likely to be included in the official figures than those involving young drivers (Elander et al., 1993).

Younger age, male gender, increased mileage, and more driving hours per week were found to be related to an increased violation score. Female gender as well as more driving hours per week related to more errors. One interesting phenomenon is that the DBQ-age correlation was dependent on the mean age of the sample. Across all samples, violations decreased with age, but in the youngest sample, violations increased with age. For errors, a different pattern was found; errors decreased for young drivers but remained relatively constant with age within the older samples. A longitudinal study of Wells et al. (2008a,b) provided more precise insight into the confounding roles of age and experience. Their results showed that for drivers with 6, 12, 24, and 36 months of experience, older drivers

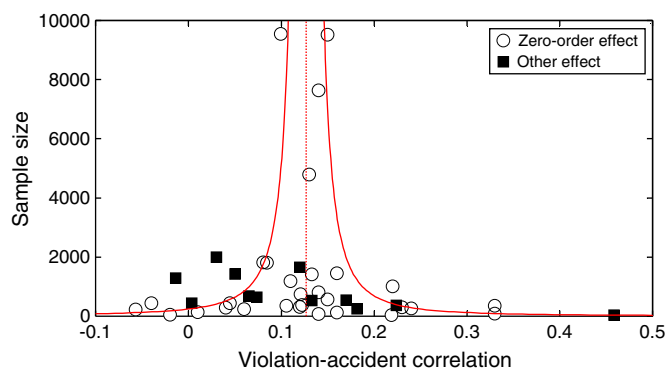


Fig. 3. Funnel plot for the violations-accident correlation. A 95% confidence interval is drawn around the overall zero-order effects.

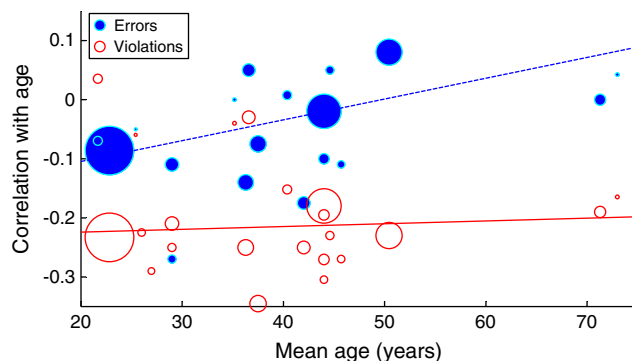


Fig. 4. DBQ errors-accident / DBQ violations-accident correlation versus mean age of the sample. Sample-size-weighted regression lines are included for errors (dashed line) and violations (solid line). Only zero-order correlations were used. The area of the dots corresponds to the sample size; the dots in the legend represent the area corresponding to a sample size of 500.

Table 2
Correlations between DBQ scores and self-reported accidents in longitudinal study.

		DBQ violations			
		6 months	12 months	24 months	36 months
Accidents	6 months	.15	.14	.14	.12
	12 months	.12	.12	.13	.14
	24 months	.10	.10	.11	.10
	36 months	.09	.09	.10	.10
		DBQ errors			
		6 months	12 months	24 months	36 months
Accidents	6 months	.09	.06	.06	.07
	12 months	.07	.07	.06	.07
	24 months	.06	.05	.08	.06
	36 months	.06	.07	.10	.08

Note. These data are based on the data archive of TRL (2008). The longitudinal study consisted of four surveys taken at 6, 12, 24, and 36 months after passing the driving test. Drivers had to complete the DBQ and indicate the number of accidents in the past period (i.e., 6 months since passing the test, 6–12 months, 12–24 months, and 24–36 months). Only participants without missing DBQ and accident values were included. Because of attrition, sample sizes varied between 9,138 (for the 6-month vs. 6-month correlation) and 1,718 (for the 36-month vs. 24-month correlation). The violations-accidents correlations were averaged from the violations scale and aggressive violations scale. The errors-accidents correlations were averaged from the errors scale, inexperience scale, and slips scale. The average violations-accident correlations as reported in the table are .13, .10, and .12 for retrospective (above-diagonal), prospective (below-diagonal), and same-questionnaire (diagonal) predictions, respectively. The average errors-accident correlations are .08, .07, and .07, respectively. All correlations are significantly different from 0, $p < .05$.

had consistently lower violation scores than the younger drivers. However, violations increased with increasing experience. In other words, violations decreased with age, but increased with experience. The effects of experience can likely be explained by the fact that novice drivers are building confidence in traffic and therefore increase violations, while at the same time they are still in the learning phase, hence increasing skill and reducing errors (see also De Winter, Wieringa, Kuipers, Mulder, & Mulder, 2007 for a driving simulator study on errors and violations during training of pre-licensed drivers). At a later age driving skill can be assumed to be relatively constant and no further reduction of errors is expected.

It is to be acknowledged that there are limitations to using self-reported data for accident prediction. As already pointed out by Reason et al. (1990), the DBQ is a powerful means to measure behaviours that are “too private to be detected by direct observation,” but at the same time the DBQ responses “are several stages removed from the actuality of what goes on behind the wheel” (pp. 1329–1330). Furthermore, asking drivers how many times they had slips is somewhat ironic: “Unconscious errors may be hard to remember precisely because they are unconscious” (Björnskau & Sagberg, 2005, p. 137). Below we discuss a number of biases that may have affected the correlations found in our meta-analysis.

4.1. Effects of data analysis method

As mentioned in the results section, differences were found between the zero-order effects reported in a tabular form, on the one hand, and the effects reported in multivariate analysis, in a table including only the significant effects, or in the text of a study, on the other. The zero-order correlations were stronger predictors of accidents, which can be explained by the fact that in multivariate analysis, the effects of two important variables, age and gender, were partialled out, attenuating the observed relationship. We recommend that researchers always report the correlation matrix with the zero-order effects, also if these are statistically insignificant, instead of reporting the results of multivariate analysis only. Meta-analysts and other researchers being interested in theory-forming will be greatly assisted by having the complete information available.

Some variability in the observed correlations may also be caused by researchers using dichotomization (e.g., Dobson et al., 1998) or data transformations (e.g., Lawton, Parker, Stradling, & Manstead, 1997; Parker, Reason, Manstead, & Stradling, 1995), by the factor score estimation method (sum scores by factor versus more refined methods, see DiStefano, Zhu, & Mindrila, 2009 for a review on factor score methods), or by the regression analysis method (logistic regression as opposed to regular least-squares regression). This inevitably had some effect on the correlations obtained. However, a study of Gebers (1998) showed that accident-prediction results were extremely similar across different methods of regression analysis. Furthermore, we tried dichotomization and various transformations on the available raw data of TRL (2008) and found that these procedures had only little effect on the DBQ-accident correlations.

4.2. Publication bias

Publication bias arises from the tendency of researchers to handle the reporting of experimental results that show a statistically significant finding differently from results that support the null hypothesis. Our meta-analysis took preventive measures against publication bias by conducting a thorough literature search and by including not only published but also unpublished documents. The funnel plots revealed a slight publication bias, although this issue was only apparent when the sample size was very small and only the significant effects were reported. The funnel plots were relatively symmetric for the zero-order correlations.

4.3. Bias due to consistency motif

Because the DBQ relies on self-reported data, it may be sensitive to several biases, such as overestimation of one's own skills (Åberg, Afram, & Nilsson, 2005) and social desirability. For different perspectives on this latter topic in relation to the DBQ see Af Wählberg (2010), Harrison (2009a), Lajunen and Summala (2003), and Sullman and Taylor (2010). However, the potentially most serious bias for accident prediction is so-called consistency motif. As rightly recognized by Af Wählberg et al. (in press), the DBQ's ability to predict accidents is exclusively based on analyses in which self-reported behaviors (i.e., the DBQ errors and violations) were correlated with self-reported accidents measured on the same occasion. Because people tend to fill in questionnaire items in a congruent manner, it is theoretically possible that the correlation between the self-reported behaviors and self-reported accidents is inflated, or even spurious.

Although the majority of DBQ studies used self-reported behaviors as criteria, there exist a number of studies that circumvented consistency motif and still yielded relatively large effects. One way to deal with consistency motif is to correlate with an objective criterion instead of self-reported behavior. The present meta-analysis showed that no definitive conclusions could be made regarding recorded accidents, but several studies used other types of objective criteria that may serve as proxy variables for accidents. For example, the DBQ violations factor was found to be significantly correlated with driving speed, gap acceptance behavior, and lane variability in a driving simulator (Charlton, 2003; De Winter, Spek, De Groot, & Wieringa, 2009; Reimer, D'Ambrosio, Coughlin, Kafirissen, & Biederman, 2006; Schwebel et al., 2006; Stephens & Groeger, 2009) and with logged speeding in real cars (.43; Wallén Warner, 2006). DBQ errors and violations have also been correlated with psychomotor performance (up to $-.35$ and $-.47$, respectively; Sümer, Ayaşık, & Er, 2005) as well as with on-road driving test performance ($-.42$ and $-.45$, respectively; Conner & Lai, 2005).

A second remediation to consistency motif is to assess correlations between the self-reported behaviors of different persons. Bianchi and Summala (2004) used DBQ scores of parents to predict accident involvement of their children, but no significant effects were found,

possibly due to their relatively small sample size. A study of Rimmö (1999) compared the number of aberrations reported by drivers and by two observers sitting in the rear of the car and found significant correlations between the driver and the observers for violations (.41 and .34 between the driver and Observer 1 and 2, respectively) and for inexperience errors (.58 and .38).

A third way to alleviate consistency motif is to make use of a sufficiently large time span between questionnaires (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). Parker, Reason, et al. (1995) and Parker, West, Stradling, and Manstead (1995) did not administer the DBQ and the questionnaire about accidents on the same occasion, but asked the participants about their accident involvement two years before and one year after filling in the DBQ. Parker, Reason, et al. (1995) reasoned that due to the long time gap, consistency motif would be minimal. In both studies did traffic accidents correlate with DBQ violations, indicating that the violations factor can predict self-reported accidents both retrospectively and prospectively. In our supplementary analysis, we showed that both violations and errors predicted past and future accidents with about equal strength, which corroborates the findings of Parker and colleagues and suggests that the effects of consistency motif are minimal.

4.4. Bias due to common scale anchors

Past research in driving simulators showed that there is a negative correlation between errors and violations (De Winter et al., 2007) and Lajunen and Summala (1995) also found a negative correlation between the skill and safety-motive dimensions on the Driver Skill Inventory. In contrast, many DBQ studies reported a markedly strong positive correlation (between .3 and .7) between errors and violations (e.g., Conner & Lai, 2005; Dobson et al., 1998; Freeman et al., 2009; Özkan & Lajunen, 2005a; Sümer, Ayvaşık, et al., 2005; Sümer, Lajunen, et al., 2005), unless the factors were rotated orthogonally so that the interfactor correlations were zero per definition (e.g., Åberg & Wallén Warner, 2008; Gabaude, Marquié, & Obriot-Claudel, 2010; Sullman et al., 2002). Exposure effects could not be the reason, considering that their correlations with errors and violations were found to be close to zero (between $-.09$ and $.17$, see Table 1).

A likely explanation for the errors-violations correlation is bias due to common scale anchors (Podsakoff et al., 2003). Some people may have a different image of the meaning of the DBQ item anchors than others. That is, the meaning of anchors such as “hardly ever,” “occasionally,” and “quite often” is subjective and sensitive to individual differences, resulting in a spurious correlation between errors and violations. This issue was also recognized by Bjørnskau and Sagberg (2005), who administered the DBQ by asking the drivers how many times during the last month the behaviors were committed in addition to using the traditional six-point scale. The results showed that their modified DBQ was able to reveal correlation between errors and violations and driving experience that were not detected using the traditional DBQ.

4.5. Bias due to measurement error

Measurement error is always present in research data, as a result of which the observed correlations are attenuated compared to the true correlations between constructs. There exist three error processes that should be taken into account when estimating the true effect sizes: random response error, transient error, and specific factor error (Schmidt & Hunter, 1999). Because errors and violations are the result of a factor analytic procedure, it can be assumed that specific factor error is small. When the same measure is correlated on two occasions, an estimate of the coefficient of stability (CS) is obtained, providing a measure of the random response error and transient error.

Based on two measurements separated by seven months, Parker, Reason, et al. (1995; see also Reason, Manstead, Stradling, Parker, & Baxter, 1991) found a CS of .69, .75, and .81 for errors, lapses, and violations, respectively. Similarly, Burgess and Webley (1999) found a CS of .59, .61, and .69 for errors, lapses, and violations, respectively, reported three months before and three months after following a rectification course. Rimmö (1999) found a CS of .75, .72, .70, and .84 for mistakes, inattention errors, inexperience errors, and violations, respectively, across a six-month interval. By using a time interval of three years, Özkan, Lajunen, and Summala (2006) found a CS of .50 for errors and of .76 for violations. Finally, Harrison (2009b) found a CS of .65, .72, .75, and .72 for errors, lapses, violations, and aggressive violations, for a six-month period. The CS of the accident data criterion is considerably lower. For the TRL (2008) data used in our supplementary analysis, we calculated the CS values for all six combinations of accident data amongst the four 6- and 12-month time periods, and found an average CS of .11.

Assuming no systematic error and a CS of .65, .75, and .11 for the DBQ errors, DBQ violations, and the accident criterion, respectively, it can be calculated with the Spearman-Brown correction-for-attenuation formula that the observed DBQ-accidents correlation is attenuated by a factor 3.75 for errors and 3.45 for violations. The true DBQ-accident correlations may therefore be around .4 for both errors and violations, which is quite respectable compared to validity coefficients in other psychometric domains. As a comparison, the true correlation between personality dimensions and job performance, an extensively studied topic, approximates .2 or .3 at best (Barrick & Mount, 1991).

5. Conclusion and recommendations

Twenty years after the seminal article of Reason et al. (1990), the DBQ has gained enormous popularity. The present meta-analysis synthesized the available information and showed that both DBQ errors and violations are significant predictors of self-reported accidents. The correlation between the DBQ factors and self-reported accidents approximates .1 and may be in the region of .4 when measurement error is corrected for. We discussed other sources of bias as well, including consistency motif and common scale anchors. Our analysis showed that the DBQ predicted accidents prospectively as well as retrospectively with comparable strength, which suggests that consistency motif does not have a negative effect on the validity of the DBQ.

This study accurately determined the overall DBQ-accident relationship in a total sample of more than 45,000 respondents. Different research designs are now recommended for developing further confidence in the DBQ. Specifically, we recommend large-sample validation studies with registered data as a criterion, such as driving simulator performance or violation information from registered speeding. Further research into effect size bias, such as the effects of measurement error and scale anchors, is recommended as well.

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